Python Implementation in Zoology

What is Python Implementation in Zoology?

Python implementation in zoology means using the Python programming language to study animals through data analysis, behavioral modeling, population tracking, wildlife monitoring, and AI-based animal research**.**

Zoologists collect data about:

- Animal movement
- Population growth
- Feeding behavior
- Habitat conditions
- Health & disease patterns

Python helps scientists analyze animal data, simulate ecosystems, monitor wildlife, and predict animal behavior efficiently.

It is widely used in:

- Wildlife conservation
- Animal behavior research
- Ecology & biodiversity studies
- Veterinary science
- Environmental zoology
- AI-based animal monitoring

Why Python is Used in Zoology

Python helps zoologists to:

- Track animal population changes
- Analyze migration patterns
- Study animal health data
- Detect species using AI & image processing
- Monitor environmental conditions
- Visualize animal research data

Program 1 — Animal Population Densit(Used in: Wildlife ecology)

```
animals = int(input("Number of animals: "))
area = float(input("Area in sq.km: "))

density = animals/area
print("Animal Population Density:",density)
```

Program 2 — Animal Weight Growth Calculator(Used in: Animal physiology)

```
initial = float(input("Initial weight (kg): "))
final = float(input("Final weight (kg): "))
days = int(input("Days: "))

growth = (final-initial)/days
print("Weight Growth per Day:",growth)
```

Program 3 — Fish Population Growth Model(Used in: Marine zoology)

```
import numpy as np
import matplotlib.pyplot as plt

t = np.linspace(0,10,100)
population = 50*np.exp(0.2*t)

plt.plot(t,population)
plt.xlabel("Time")
plt.ylabel("Fish Population")
plt.show()
```

Program 4 — Animal DNA GC Content(Used in: Animal genetics)

```
dna = input("Enter Animal DNA Sequence: ").upper()

gc = dna.count('G') + dna.count('C')
percent = gc/len(dna)*100

print("GC Content:",percent,"%")
```

Program 5 — Predator–Prey Simulation(Used in: Ecology & food chain studies)

```python
import numpy as np
import matplotlib.pyplot as plt

t = np.linspace(0,20,200)
prey = 40*np.exp(0.1*t)
predator = 9*np.exp(0.05*t)

plt.plot(t,prey,label="Prey")
plt.plot(t,predator,label="Predator")
plt.legend()
plt.show()
```

Program 6 — Animal Disease Risk Predictor (Used in: Veterinary zoology)

```python
from sklearn.tree import DecisionTreeClassifier

temp = [[30],[20],[35],[18]]
disease = [1,0,1,0]

model = DecisionTreeClassifier()
model.fit(temp,disease)

print("Disease Prediction:",model.predict([[32]])[0])
```

Program 7 — Animal DNA Translation(Used in: Used in molecular zoology & genetics)

```
codon_table = {
'ATA':'I','ATC':'I','ATT':'I','ATG':'M',
'ACA':'T','ACC':'T','ACG':'T','ACT':'T',
'AAC':'N','AAT':'N','AAA':'K','AAG':'K',
'AGC':'S','AGT':'S','AGA':'R','AGG':'R',
'CTA':'L','CTC':'L','CTG':'L','CTT':'L',
'CCA':'P','CCC':'P','CCG':'P','CCT':'P',
'CAC':'H','CAT':'H','CAA':'Q','CAG':'Q',
'CGA':'R','CGC':'R','CGG':'R','CGT':'R',
'GTA':'V','GTC':'V','GTG':'V','GTT':'V',
'GCA':'A','GCC':'A','GCG':'A','GCT':'A',
'GAC':'D','GAT':'D','GAA':'E','GAG':'E',
'GGA':'G','GGC':'G','GGG':'G','GGT':'G',
'TCA':'S','TCC':'S','TCG':'S','TCT':'S',
'TTC':'F','TTT':'F','TTA':'L','TTG':'L',
'TAC':'Y','TAT':'Y','TAA':'_','TAG':'_',
'TGC':'C','TGT':'C','TGA':'_','TGG':'W'
}

dna = input("Enter Animal DNA: ").upper()
protein = ""

for i in range(0,len(dna),3):
    codon = dna[i:i+3]
    protein += codon_table.get(codon,"")

print("Protein Sequence:",protein)
```

Program 8AnimalMovementSpeedAnalysis(Usedin:Animaltrackingresearch,Behavioral zoology)

```
import numpy as np

distance = np.array([5,10,15,20])
time = np.array([1,2,3,4])

speed = distance/time

print("Animal Movement Speed:",speed)
print("Average Speed:",np.mean(speed))
```

Program 9 — Marine Biodiversity Index (Simpson Index Used in:,Marine zoology,Ecosystem analysis)

```
species = [20,30,25,25]

N = sum(species)
D = sum([n*(n-1) for n in species])/(N*(N-1))

print("Simpson Biodiversity Index:",1-D)
```

Program 10 — Logistic Animal Population Model

```
import numpy as np
import matplotlib.pyplot as plt

t = np.linspace(0,20,100)
K = 500
r = 0.2
P0 = 50

P = K/(1+((K-P0)/P0)*np.exp(-r*t))

plt.plot(t,P)
plt.xlabel("Time")
plt.ylabel("Animal Population")
plt.show()
```

Program 11 — Animal Classification using Machine Learning

```
from sklearn.neighbors import KNeighborsClassifier

data = [[2],[4],[6],[8]]
label = ["Small","Medium","Large","Very Large"]

model = KNeighborsClassifier(n_neighbors=1)
model.fit(data,label)

print("Animal Size Class:",model.predict([[5]])[0])
```

ZOOLOGY MINI PROJECT USING PYTHON

Project Title:

Predator–Prey Ecosystem Simulation

```
import numpy as np
import matplotlib.pyplot as plt

# ---------------- USER INPUT ----------------
prey = float(input("Initial Prey Population: "))
predator = float(input("Initial Predator Population: "))

prey_birth = float(input("Prey Birth Rate (e.g.,0.1): "))
predation_rate = float(input("Predation Rate (e.g.,0.02): "))
predator_growth = float(input("Predator Growth Rate (e.g.,0.01): "))
predator_death = float(input("Predator Death Rate (e.g.,0.1): "))
```

```
# ---------------- SIMULATION ----------------
time_steps = 100

prey_list = []
predator_list = []

for t in range(time_steps):
    prey_growth = prey_birth * prey
    prey_loss = predation_rate * prey * predator

    predator_gain = predator_growth * prey * predator
    predator_loss = predator_death * predator

    prey = prey + prey_growth - prey_loss
    predator = predator + predator_gain - predator_loss

    if prey < 0:
        prey = 0
    if predator < 0:
        predator = 0

    prey_list.append(prey)
    predator_list.append(predator)
```

```python
# ---------------- EXTINCTION CHECK ----------------
if prey == 0:
    print("⚠ Prey Species Extinct")
if predator == 0:
    print("⚠ Predator Species Extinct")

# ---------------- PREDICTION ----------------
print("Final Prey Population:",prey)
print("Final Predator Population:",predator)

# ---------------- GRAPH ----------------
plt.plot(prey_list,label="Prey Population")
plt.plot(predator_list,label="Predator Population")

plt.xlabel("Time Steps")
plt.ylabel("Population")
plt.title("Advanced Predator-Prey Ecosystem Simulation")

plt.legend()
plt.show()
```